Lecture 11 // Sept 29 // Instanton Floer Homology

The rest of the course will follow this general outline.

1. Broad overview of Instanton Floer Homology
2. Formal properties & applications
3. More detailed discussion of (1)
4. More recent papers

Instanton Floer Homology is a 3-manifold invariant. Originally, it was defined only for integer homology spheres, and we'll restrict to Y a $\mathbb{Z}\mathrm{HS}^3$ for simplicity, but the constructions we'll discuss can be extended.

The general procedure is as follows:

$$Y \rightsquigarrow C(Y) \rightsquigarrow H_*(C(Y)),$$

$$\mathbb{Z}\mathrm{HS}^3 \qquad \text{Instanton chain complex} \qquad \text{Instanton Floer homology.}$$

Where does $C(Y)$ come from? Two slogans:

1. ∞-dimensional Morse homology associated to the CS functional, i.e., there is a mysterious S^1-valued functional on an auxiliary space associated to Y (like $\mathcal{A}/\mathcal{G}$, but for Y^3):

   $$\mathrm{CS} : \mathcal{A}/\mathcal{G} \longrightarrow \mathbb{R}/\mathbb{Z}.$$

   Whenever you have an $\mathbb{R}$-valued function, we take the Morse homology:

   - the critical points of CS become the generators of the chain complex,
   - the differentials are defined (roughly) by studying the gradient flow of CS.

2. The generators of $C(Y)$ are in correspondence with conjugacy classes of $\mathrm{SU}(2)$ representations of $\pi_1(Y)$. So $C(Y)$ actually comes from π_1.

Today, we'll broadly discuss (1) & (2), and why they're the same.

What is the Chern-Simons Functional

Take Y^3 a $\mathbb{Z}\mathrm{HS}^3$ (though nothing today depends on that). Fix an $\mathrm{SU}(2)$ bundle over Y. Since $c_2(Y) = 0$, there's only one such bundle, the trivial one $E = Y \times \mathbb{C}^2 \to Y$.

Just like before, we study $\mathcal{A} = \{\mathrm{SU}(2) \text{ connections on } E\}$ and $\mathcal{G} \curvearrowright \mathcal{A}$, $\mathcal{G} = \Gamma(\mathrm{Aut}(E))$. Since E is trivial, $\mathcal{G} \cong \mathrm{Maps}(Y, \mathrm{SU}(2))$.

Note that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$, since $\mathrm{SU}(2) \cong S^3$, and maps $Y \to S^3$ are determined by the degree $\deg(g) \in \mathbb{Z}$.

$\mathcal{A}/\mathcal{G}$ = configuration space / gauge, the domain of CS.

Definition 1

The Chern-Simons Functional is a map

$$\mathrm{CS} : \mathcal{A}/\mathcal{G} \longrightarrow S^1, \qquad \mathrm{CS}(A) = \frac{1}{8\pi^2}\int_Y \mathrm{tr}\left(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\right).$$

By this, we mean first: fix a global trivialization of E. In this global trivialization, ∇ is represented by a matrix of 1-forms A, which is the input of the integral.

Exercise 2

If $g \in \mathcal{G}$ is a global gauge transformation, $\mathrm{CS}(g^*A) = \mathrm{CS}(A) + \deg(g)$.

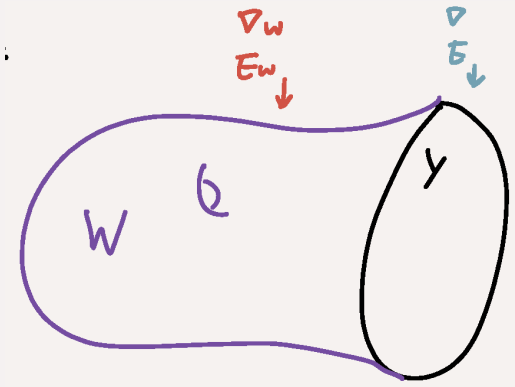
This gives another way of saying that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$; each connected component corresponds to a trivialization, so this is saying that, up to homotopy, there are $\mathbb{Z}$-many trivializations of E. The value of CS depends on which trivialization you choose. This definition is super explicit but not as insightful as the following alternate definition.

Definition 3

Let A be a connection on E. Choose any:

- smooth 4-manifold W^4 such that $\partial W \cong Y$,
- SU(2)-bundle E_W on W such that $E_W|_Y \cong E$,
- SU(2)-connection ∇_W such that $\nabla_W|_Y = \nabla$.

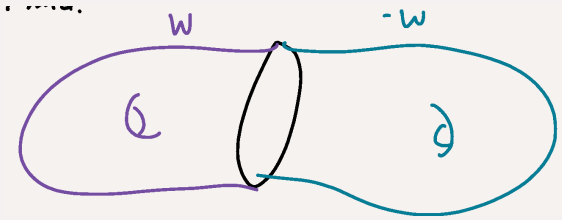
Then $\mathrm{CS}(\nabla) = \frac{1}{8\pi^2}\int_W \mathrm{tr}(F_{\nabla_W} \wedge F_{\nabla_W})$.



This definition follows from the definition of relative characteristic classes. But as-is, you should be skeptical that this is well defined. Let's choose a different W', $E_{W'}$, ∇'. Then

$$\mathrm{CS}(\nabla) = \frac{1}{8\pi^2}\int_W \mathrm{tr}(F_{\nabla_W} \wedge F_{\nabla_W}) \quad \text{and} \quad \mathrm{CS}(\nabla) = \frac{1}{8\pi^2}\int_{W'} \mathrm{tr}(F_{\nabla_{W'}} \wedge F_{\nabla_{W'}}).$$

If this definition is well-defined, then $\mathrm{CS}(\nabla) - \mathrm{CS}(\nabla') = 0$. To compute the difference, glue W and $-W'$ along Y to get an SU(2) bundle over a closed 4-manifold.



Then their difference

$$\frac{1}{8\pi^2}\int_W \operatorname{tr}(F_{\nabla_W}\wedge F_{\nabla_W}) - \frac{1}{8\pi^2}\int_W \operatorname{tr}(F_{\nabla_{W'}}\wedge F_{\nabla_{W'}}) = \left\langle c_2(E_\Sigma), [W\cup_Y -W']\right\rangle \in \mathbb{Z}.$$

Thus different W's change $\mathrm{CS}(\nabla)$ by an integer. Thus $\mathrm{CS}(\nabla)$ is well-defined in S^1.

Remark 4

The bundles $E_W, E_{W'}$ over W and W' glue together to a well-defined bundle E_Σ over $W\cup_Y -W'$. The same is true for ∇_W and $\nabla_{W'}$, obtaining ∇_Σ. The Chern-Simons integral of this connection is that of a closed 4-manifold.

$$\mathrm{CS}(\nabla_\Sigma) = \frac{1}{8\pi^2}\int_{W\cup_Y -W'} \operatorname{tr}(F_{\nabla_\Sigma}\wedge F_{\nabla_\Sigma}) \in \mathbb{Z},$$

and splits as $\displaystyle\int_W \operatorname{tr}(F_{\nabla_\Sigma}|_W \wedge F_{\nabla_\Sigma}|_W) - \int_{W'} \operatorname{tr}(F_{\nabla_\Sigma}|_{W'} \wedge F_{\nabla_\Sigma}|_{W'})$.

However, for the connections ∇_W and $\nabla_{W'}$ over 4-manifolds with non-empty boundary, $\mathrm{CS}(\nabla_W)\notin\mathbb{Z}$. In our proof of well-definedness, we relied on some gluing procedure without verifying what that procedure actually is: this was incredibly sloppy. We should also ask whether this definition plays well with $\mathcal{G}$.

In order to glue $E_W \& E_{W'}$ together, we must fix identifications $E_W|_Y \cong E_{W'}|_Y$. Since there is only one $\mathrm{SU}(2)$ bundle over Y, the trivial one, we could argue that $E_W, E_{W'}$ are trivial bundles, thus E_Σ is trivial, so the difference is zero. While the first part of this is true, like we saw with $\mathbb{CP}^1$, two trivial bundles could be glued via some nontrivial boundary map, like $\mathbb{Z}^n$, which results in a nontrivial bundle. Even simpler, two trivial $\mathbb{R}$-bundles over the interval I could be glued via a twist, resulting in the Möbius band. The number of these different maps is the number of retrivializations of the bundle on the boundary. So we must ask: how many retrivializations of ∇_W are there? The answer is $\mathbb{Z}$, as we saw earlier, and these $\mathbb{Z}$-many gluings give $\mathbb{Z}$-many $\mathrm{SU}(2)$ bundles over W!

Claim 5. *Re-trivializing a bundle E_W over $\partial W\cong Y$ by composing with some $g\in\mathcal{G}$ changes $c_2(E_\Sigma)$ by* $\deg g\in\mathbb{Z}$.

Lemma 6

The two definitions agree.

Proof. This follows immediately from Stokes' theorem: $\left(\int_W d\beta = \int_{\partial W} \beta\right)$.

$$8\pi^2 \operatorname{CS}(A) = \int_Y \operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right) = \int_W d\left(\operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)\right).$$

Simplifying,

$$d\left(\operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)\right) = \operatorname{tr}\left(d(A \wedge dA) + \frac{2}{3} d(A \wedge A \wedge A)\right)$$

$$= \operatorname{tr}\left(dA \wedge dA - A \wedge d^2 A + \frac{2}{3}\left(dA \wedge A \wedge A - A \wedge dA \wedge A + A \wedge A \wedge dA\right)\right)$$

$$= \operatorname{tr}\left(dA \wedge dA + \frac{2}{3} dA \wedge A \wedge A - \frac{2}{3} A \wedge dA \wedge A + \frac{2}{3} A \wedge A \wedge dA\right)$$

$$= \operatorname{tr}(dA \wedge dA + 2\, dA \wedge A \wedge A) = \operatorname{tr}(dA \wedge dA + dA \wedge A \wedge A + A \wedge A \wedge dA)$$

$$= \operatorname{tr}\left((dA + A \wedge A) \wedge (dA + A \wedge A)\right) + \operatorname{tr}(A \wedge A \wedge A \wedge A) = \operatorname{tr}(F_A \wedge F_A),$$

since $\operatorname{tr}(A \wedge A \wedge A \wedge A) = 0$. Thus

$$\int_Y \operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right) = \int_W \operatorname{tr}(F_A \wedge F_A) = 8\pi^2 \left\langle c_2(E_\Sigma), [W \cup_Y -W'] \right\rangle. \qquad \square$$

Remark 7
Timeline: this all happened pre-Donaldson, post relative characteristic classes.

Just like in Donaldson's theorem, we want to study the critical points of CS, which we'll denote by $\operatorname{Crit}(\mathrm{CS}) \subset \mathcal{A}/\mathcal{G}$. What about interpretation #2?

Theorem 8
$\nabla \in \mathcal{A}/\mathcal{G}$ is a critical point of the Chern-Simons functional iff ∇ is flat, i.e., $F_A = 0$.

Proof.

$$\nabla \in \operatorname{Crit}(\mathrm{CS}) \iff \left.\frac{d}{dt}\right|_{t=0} \operatorname{CS}(\nabla + tb) = 0 \iff \left.\frac{d}{dt}\right|_{t=0} 8\pi^2 \operatorname{CS}(\nabla + tb) = 0$$

$$\iff \int_Y \operatorname{tr}\left((A + tb) \wedge d(A + tB) + \frac{2}{3}\big((A + tb) \wedge (A + tb) \wedge (A + tb)\big)\right) = 0$$

Using $d(A + tB) = dA \wedge B - A \wedge dB$,

$$\iff \int_Y \operatorname{tr}\left(B \wedge dA + A \wedge dB + \frac{2}{3} B \wedge A \wedge A + \frac{2}{3} A \wedge B \wedge A + \frac{2}{3} A \wedge A \wedge B\right) = 0$$

$$\iff \int_Y \operatorname{tr}(2B \wedge dA + 2B \wedge A \wedge A) = 0 = 2\int_Y \operatorname{tr}(B \wedge F_A) = 0 \quad \forall B$$

$$\iff F_A = 0.$$

$$\operatorname{grad}_A \mathrm{CS} = *F_A. \qquad \square$$

Theorem 9
Critical points of the Chern-Simons functional modulo gauge are in bijection with SU(2) representations of $\pi_1(Y)$ modulo conjugation.

Definition 10
The character variety of a 3-manifold is

$$R(Y) = \{\text{SU(2) representations of } \pi_1(Y)\}/\sim,$$

where $\rho_0 \sim \rho_1$ iff $\rho_1 = g^{-1} \cdot \rho_0 \cdot g$.

In this language, $\mathrm{Crit}(\mathrm{CS}) \leftrightarrow R(Y)$. We now prove the theorem.

Proof. Taking the holonomy gives a map

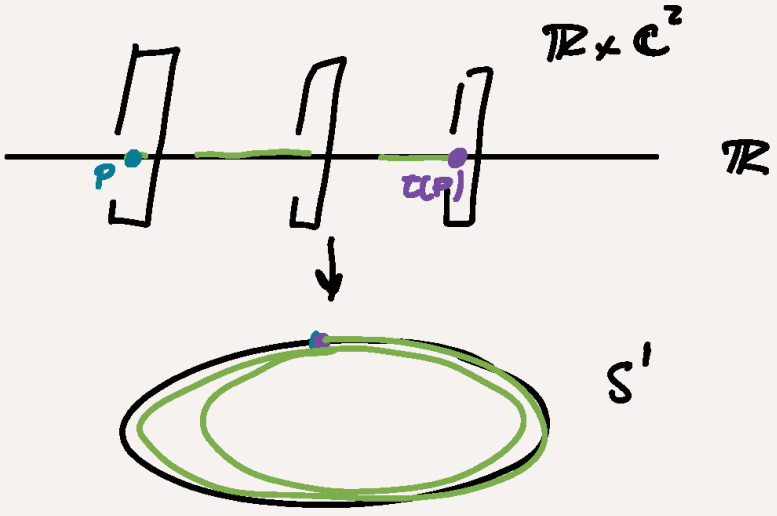
$$\{\text{flat connections}\} \to \{\text{SU(2) representations of } \pi_1(Y)\}.$$

Since $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$, this holonomy descends to a map on the quotient

$$\{\text{flat connections}\}/\mathcal{G} \to \{\text{SU(2) representations of } \pi_1(Y)\}/\sim = R.$$

It's not obvious this map is surjective. To show surjectivity, let $\rho : \pi_1(Y) \to \mathrm{SU}(2)$ be a representation of $\pi_1(Y)$. Construct an SU(2) bundle E and connection ∇ on E as follows:

1. Form $\widetilde{E} = \widetilde{Y} \times \mathbb{C}^2$, the trivial bundle over the universal cover of Y.
2. Quotient out $\widetilde{E}$ by identifying the fibres $p \times \mathbb{C}^2$ and $\tau(p) \times \mathbb{C}^2$ for $\tau \in \mathrm{Deck}(\widetilde{Y}) \cong \pi_1(Y)$ via the matrix given by $\rho(\gamma) \in \mathrm{SU}(2)$, where $\gamma \in \pi_1(Y) \leftrightarrow \tau$ between $p, \tau(p)$ (where γ lifts to τ).



This defines a bundle E over Y, as well as parallel transport on E by pushing down parallel transport on $\widetilde{E}$. By construction, this parallel transport will have holonomy ρ. This shows surjectivity. $\square$